Additional Problems for Ch 8 HRW – PE and conservation of Energy
Addisionele Probleme vir Hfst 8 HRW –PE en behoud van energie

	9^{th} Edition	8^{th} Edition
Questions	3, 6, 7	1, 6, 7
Problems	6, 15, 23, 28, 31, 33, 55, 57, 62, 98	8, 9, 21, 30, 29, 33, 59, 53, 64, 98

6	8	(a) $W_g = 0.15$ J. (b) $W_g = 0.094$ J. (c) $U = 0.16$ J. (d) $U = 0.031$ J. (e) $U = 0.063$ J. (f) The new information $(v_i \neq 0)$ is not involved in any of the preceding computations; the above results are unchanged.	(a) $W_g = 0.15$ J. (b) $W_g = 0.11$ J. (c) $U = 0.19$ J (d) $U = 0.038$ J (e) $U = 0.075$ J (f) The new information $(v_i \neq 0)$ is not involved in any of the preceding computations; the above results are unchanged.
15	9	v = 130(1000/3600) = 36.1 m/s. (a) L = 257 m.≈ 2.6×10^2 m (b) The answers do not depend on the mass of the truck. They remain the same if the mass is reduced. (c) If the speed is decreased, h and L both decrease (note that h is proportional to the square of the speed and that L is proportional to h).	
23	21	(a) With L = 1.20 m, $v = 4.85$ m/s. (b) $v_b = \sqrt{2gL - 2g(2r)} = 2.42$ m/s.	
28	30	k = 10 N/m (a) v = 2.8 m/s for m = 0.0038 kg and x = 0.055 m. (b) v = 2.6 m/s.	k = 10 N/m (a) v = 2.8 m/s for m = 0.0038 kg and x = 0.055 m. (b) v = 2.7 m/s.
31	29	We use SI units, so k = 1960 N/m and x_0 = –0.200 m. (a) The elastic potential energy is $\frac{1}{2}kx_0^2 = 39.2$ J. (b) $\Delta U_g = U_g$ = 39.2 J. (c) $d = h/\sin 30°$ = 4.00 m.	
33	33	(a)v_2 = 2.40 m/s	(b) $v_3 = 4.19$ m/s.
55	59	(a) $W_s = -\frac{1}{2}kx^2 = -0.90$ J. (c) $v_i = \sqrt{2K_i/m} = 1.0$ m/s.	(b) ΔE_{th} = 0.46 J
57	53	D = 1.2 m	
62	64	$v_B = 3.7$ m/s.	$v_B = 3.5$ m/s.
98	98	P = 181 W.	